AI-Guided Big Data Deployment: MRS via KooCLI



HUAWEI

Contents

1. **Disclaimer**
2. **Your Big Data Scenario**
3. **AI-Guided Process**
4. **Expected Results**
5. **Why AI-Assisted**

AI-Assisted Migration: Usage Guidelines

Practical guidance, not a formal delivery commitment

How to work with AI for VM-to-ECS migration

How to Use This Guide

This guide shows what to ask your AI assistant, what knowledge it needs, and how to verify its work to prevent errors. AI accelerates migration planning and execution -- but does not replace expert review or customer decisions.

1. What AI Does

Planning / Analysis / Automation

- AI plans, analyzes, generates scripts, and assesses risks
- It does not replace design, expert review, testing, or decisions
- Review all outputs before acting on them

2. What to Provide

Anonymized Data Only

- Use simulated, test, or anonymized data only
- Never enter real credentials, passwords, keys, or customer data
- Respect security, compliance, and confidentiality

3. How to Work with AI

Review / Authorize / Verify

- Review every AI output before acting
- Authorize each migration step explicitly
- Keep a human in the loop for all production changes

4. When to Be Cautious

Sensitive Systems / Manual Review

- For sensitive systems (financial, defense, classified), consider manual migration
- Use strict AI oversight for compliance-heavy workloads
- Always prioritize security and sovereignty requirements

AI-assisted migration improves efficiency; it does not replace human judgment. Verify every step against real data.

Your Big Data Deployment Scenario

Your Starting Point

You need to process and analyze large volumes of structured and unstructured data, but running Hadoop and Spark yourself means heavy setup, patching, and manual tuning.



Your Goal

Stand up a production-ready MRS cluster with the right node mix (Master, Core, Task) and components (HDFS, Hive, Spark, HBase). Scale up and down without downtime.



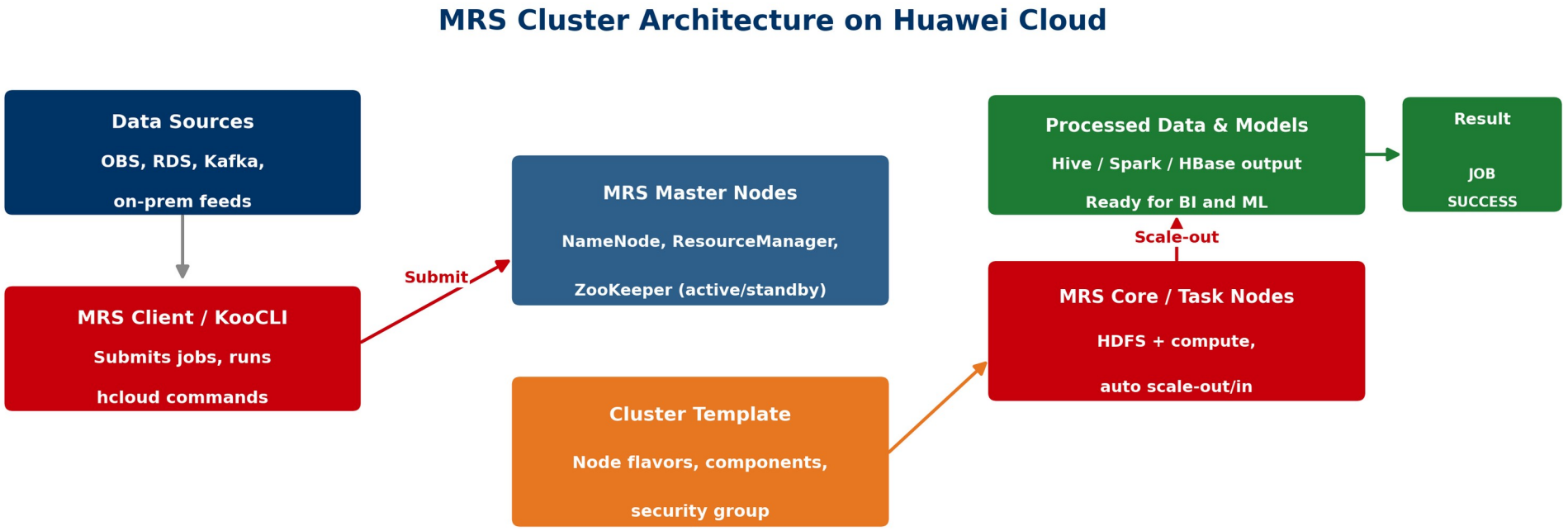
Why AI-Assisted

AI can recommend cluster topology and component sets in minutes, generate the KooCLI commands to provision it, and adjust node counts as workloads change. What used to take days of capacity planning now takes hours.



HUAWEI

Step 1: Ask AI to Discover Your Data & Workload



Example prompt: "Review my data sources and workload and recommend an MRS cluster topology -- node types, counts, and components"

AI Should Know: How to read data volumes and job patterns from your OBS and RDS sources, and how to map them to Master, Core, and Task node sizing

Source: Huawei Cloud MRS Documentation (https://support.huaweicloud.com/mrs/) — Cluster Planning Guide. Accessed: Aug 2026.

Verify: Compare the AI's topology against your actual workload. Make sure component choices (Hive, Spark, HBase, Flink) match what your jobs need

Prevent Hallucinations: The AI should reference real node flavors and component versions supported by MRS, not made-up

HUAWEI

Step 2: Ask AI to Prepare Cluster Templates

What to Ask AI

Example prompt: "Prepare an MRS cluster template with Master, Core, and Task nodes sized for our workload. Set up networking, security groups, and cross-AZ placement."



AI Should Know & Verify

The available node flavors and component combinations on Huawei Cloud MRS, how to size Core vs. Task nodes for storage versus elastic compute, and how to configure networks securely.
Verify: Each planned node type is properly sized -- not too small, not wastefully large. AI should use real flavors from the catalog, not guesses.



HUAWEI

Step 3: Ask AI to Execute Deployment

MRS Cluster Deployment: Two-Phase Process

Phase 1: Provision & Bootstrap

- KooCLI creates Master, Core, and Task nodes
- Non-disruptive — existing clusters keep running
- Bootstrap installs Hadoop, Spark, Hive, HBase
- Typical provisioning: 10–15 min for a mid-size cluster
- Monitor: node status, component install %, health

Ready →

Phase 2: Scale & Activate

- Auto scale-out adds Task nodes under load
- Component services start and register with Manager
- KooCLI verifies HDFS, YARN, job submission endpoints
- Cluster becomes available for job submission
- Roll back automatically if a component fails to start



On failure: review logs → fix issue → retry from last checkpoint

Source: Huawei Cloud MRS Documentation (https://support.huaweicloud.com/mrs/) — Cluster Deployment Guide. Accessed: Aug 2026.

HUAWEI

Step 4: Ask AI to Scale and Handle Errors

What to Ask AI

Example prompt: "Scale out Task nodes for the nightly batch window and scale them back in afterward. Handle any node or component errors automatically."



AI Should Know

Common cluster issues and how to fix them. When a node won't join the cluster one way, try an alternative approach. How to pause and resume scaling if needed.



Verify & Prevent Hallucinations

When the AI says an error is 'fixed', verify it yourself. The AI should explain what went wrong and what it did, not just say 'resolved'. Never assume a retry succeeded without checking.



HUAWEI

Step 5: Ask AI to Validate Every Cluster

What to Ask AI

Example prompt: "Check the cluster -- are all components healthy? Is HDFS balanced? Are jobs completing successfully? Give me a report with specific results for each component."



AI Should Know

What a healthy cluster looks like, how to compare metrics before and after scaling, and what performance indicators matter for your jobs.



Verify & Prevent Hallucinations

Don't accept 'everything looks good'. Ask for proof: component status, job logs, HDFS balance report. Compare performance with your previous cluster size. A human must approve before going into production.



HUAWEI

How to Make Your AI a Deployment Expert

Teach It the Process

Tell your AI what success looks like: which components to install, acceptable provisioning time, and validation criteria. Clear instructions produce better results.



Give It Real Data

Connect your AI to your Huawei Cloud account and MRS documentation. It should know your actual node flavors, networks, and component options -- not guess or assume.



Set Clear Expectations

Install deployment skills from our GitHub repository. Point your AI to Huawei Cloud MRS and KooCLI documentation. Works with OpenCode, Hermes, or Claude Code.



HUAWEI

Why AI-Assisted Deployment

Speed & Efficiency

Planning that used to take days now takes hours. AI sizes clusters and recommends components in minutes. 40% faster provisioning with automated orchestration.



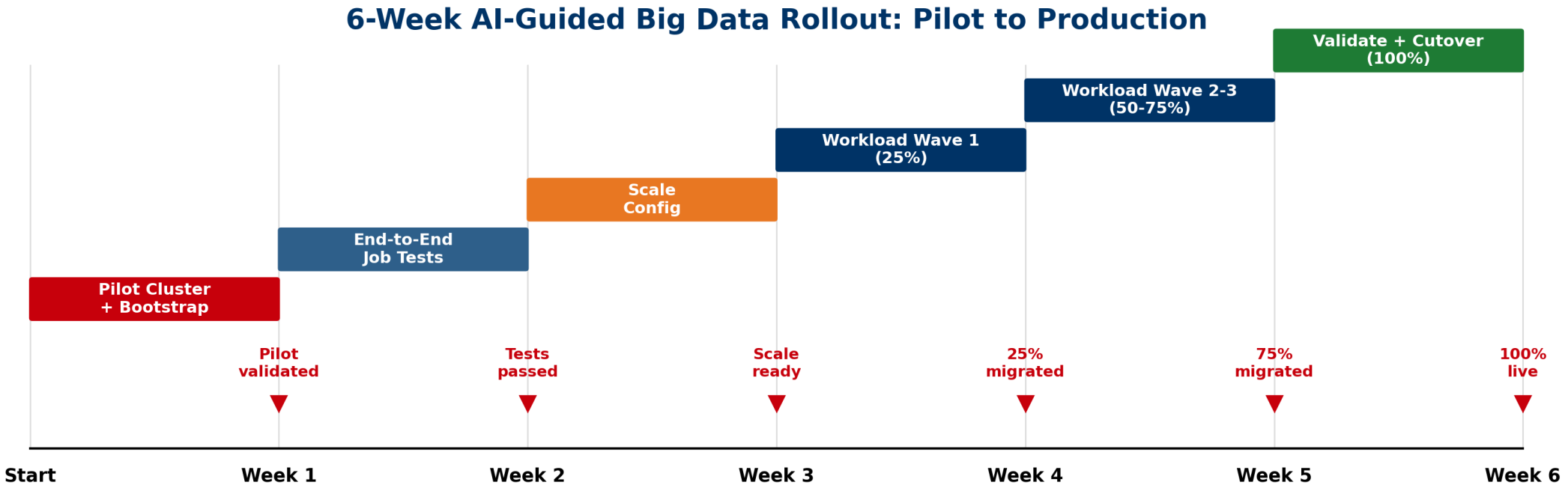
Risk Reduction

AI spots undersized nodes and missing components early. If a node won't join one way, it automatically tries another. Over 95% succeed on the first try.



HUAWEI

Your 6-Week AI-Guided Big Data Rollout



Week 1 -- Example prompt: "Set up a pilot MRS cluster and bootstrap it with our core components. Confirm everything works."
Week 2 -- Example prompt: "Run our critical Hive and Spark jobs end-to-end on the pilot cluster. Fix any issues."
Week 3-5 -- Example prompt: "Migrate workloads in waves -- 25% at a time, by job criticality. Verify each wave before proceeding."
Week 6 -- Example prompt: "Run final checks. Point all pipelines to the new cluster. Decommission the pilot."

Source: Huawei Cloud MRS Best Practices (https://support.huaweicloud.com/mrs/) — Cluster Rollout Guide. Accessed: Aug 2026.

Equip Your AI Assistant

Install Deployment Skills:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/MRS-KooCLI-Deployment

AI Tools: OpenCode, Hermes, or Claude Code

Getting Started:
1. Install skills in ~/.opencode/skills/
2. Open your AI coding tool
3. Example prompt: "Help me deploy an MRS big data cluster on Huawei Cloud using KooCLI"

Skills include: step-by-step guides, automation scripts, and runbooks

HUAWEI

Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

Copyright©2018 Huawei Technologies Co., Ltd.
All Rights Reserved.

The information in this document may contain predictive statements including, without limitation, statements regarding the future financial and operating results, future product portfolio, new technology, etc. There are a number of factors that could cause actual results and developments to differ materially from those expressed or implied in the predictive statements. Therefore, such information is provided for reference purpose only and constitutes neither an offer nor an acceptance. Huawei may change the information at any time without notice.

